MIL-S-38714B(USAF)
15 MAY 1987

Superseding
MIL-P-38714A (USAF)
14 March 1973

MILITARY SPECIFICATION

SEALANT CARTRIDGE FOR TWO COMPONENT MATERIALS

This specification is approved for use within
the Department of the Air Force and is available
for use by all Departments and Agencies of the
Department of Defense

1. SCOPE

1.1 This specification covers cartridges which are used to store, mix and apply sealants, adhesives, and similar materials.

1.2 Classification. The cartridges shall be of six types.

Type I - Injection type cartridge with standard plunger, for materials with 0.5 to 12% catalyst by weight of base resin component, for use with extrusion guns.
Size A - 6 fluid ounce (180 milliliter) cartridge
Size B - 2 1/2 fluid ounce (75 milliliter) cartridge

Type II - Barrier type cartridge with standard plunger, for materials with 12 to 50% catalyst by weight of base resin component, for use with extrusion guns.
Size A - 6 fluid ounce (180 milliliter) cartridge
Size B - 2 1/2 fluid ounce (75 milliliter) cartridge

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Warner Robins ALC/MMIRFW, Robins AFB GA 31098, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8030

AMSC N/A.

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Type III - Injection type cartridge with special plunger, for materials with 0.5 to 12% catalyst by weight of base resin component, for use with injection guns.
Size A - 6 fluid ounce (180 milliliter) cartridge

Type IV - Barrier type cartridge with special plunger, for materials with 12 to 50% catalyst by weight of base resin component, for use with injection guns.
Size A - 6 fluid ounce (180 milliliter) cartridge

Type V - Cartridge with standard plunger, front cap and rear cap only for fill and immediate use by the user.
Size A - 6 fluid ounce (180 milliliter) cartridge
Size B - 2 1/2 fluid ounce (75 milliliter) cartridge

Type VI - Cartridge with special plunger, front cap and rear cap only for fill and immediate use by the user.
Size A - 6 fluid ounce (180 milliliter) cartridge
Size B - 2 1/2 fluid ounce (75 milliliter) cartridge

1.3 Part Number. The following part numbering system is intended for use in cataloging data systems for definitive identification of the item within the ADP word length limit assigned for the purpose and other applications as appropriate.

Basic Identifier	Type	Size
MIL-S-38714-	I	2 -- 2 1/2 fl. oz.
	II	6 -- 6 fl. oz.
	III	
	IV	
	V	
	VI	

Examples:

MIL-S-38714-I 2
filled injection cartridge for extrusion gun, 2 1/2 fl. oz.

MIL-S-38714-III 6
filled injection cartridge for injection gun, 6 fl. oz.

MIL-S-38714-VI 2
empty cartridge with special plunger (ref: Semco P/N 250-PWP) and caps, 2 1/2 fl. oz.

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2. APPLICABLE DOCUMENTS

2.1 GOVERNMENT DOCUMENTS

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks, form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Federal

L-P-390	Plastic, Molding and Extrusion Material, Polyethylene and Copolymers (Low, Medium and High Density)
PPP-B-636	Boxes, Shipping, Fiberboard

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Cartridge and Kit. The requirements for the cartridge are covered in paragraphs 3.2 through 3.2.4. The requirements for the filled cartridge kits are covered in paragraphs 3.3 through end.

3.2 Cartridge. The cartridge is a unit container and assembly designed to separate the catalyst or accelerator and base component in such a manner that the two materials can be uniformly blended just prior to use.

3.2.1 Type I Cartridge. The Type I Cartridge consists of the following components:

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a. A tubular dasher rod with piston seal and valve (for containing the catalyst until used). The dasher rod shall be made of appropriate material suitable for the intended catalyst, e.g., polyethylene dasher rods for dichromate catalysts and nylon dasher rods for lead and manganese catalysts. A polyethylene dasher rod shall conform to Federal Specification L-P-390, Type I, Class H, Grade 2, Category 3. The dasher rod valve shall conform to Federal Specification L-P-390, Type I, Class L, Grade 2, Category 2. A nylon dasher rod shall have a melting point of 200°C-225°C and a specific gravity of 1.05 - 1.20. Unless otherwise specified, the color shall be natural. (See Fig. 1)

b. A dasher which will adequately mix the combined catalyst and sealant and is detachable from dasher rod.

c. A cartridge to contain base compound. The cartridge material shall conform to Federal Specification L-P-390 Type I, Class H, Grade 2, Category 3.

d. A wiper plunger, to seal the cartridge and extrude the mixed material and which is adaptable to filleting guns, (reference item: Semco P/N 250 filleting gun).

e. A cartridge seal cap to exclude foreign materials.

f. A ramrod to force inject the catalyst into the base component. The ramrod shall fit snugly in the dasher rod to preclude loss of dasher rod contents.

3.2.2 Type II Cartridge. The Type II Cartridge consists of the following components:

a. A dasher rod with detachable dasher. A polyethylene dasher rod shall conform to Federal Specification L-P-390, Type I, Class H, Grade 2, Category 3. The dasher rod valve shall conform to Federal Specification L-P-390, Type I, Class L, Grade 2, Category 2. A nylon dasher rod shall have a melting point of 200°C-225°C and a specific gravity of 1.05 - 1.20. Unless otherwise specified, the color shall be natural. (See Fig. 2)

b. A cartridge to contain the catalyst and resin components. The cartridge shall conform to Federal Specification L-P-390, Type I, Class H, Grade 2, Category 3.

c. A barrier material to separate the two components until mixed.

d. A wiper plunger, to seal the cartridge and extrude the mixed material and which is adaptable to filleting guns (reference item: Semco P/N 250 filleting gun).

e. A cartridge seal cap to exclude foreign matter.

f. A clinch band.

3.2.3 Type III Cartridge. The Type III Cartridge shall be as specified for the Type I Cartridge with the addition of a plunger with threaded plug (reference item: Semco P/N 250 PWP) for adaptation to injection guns.

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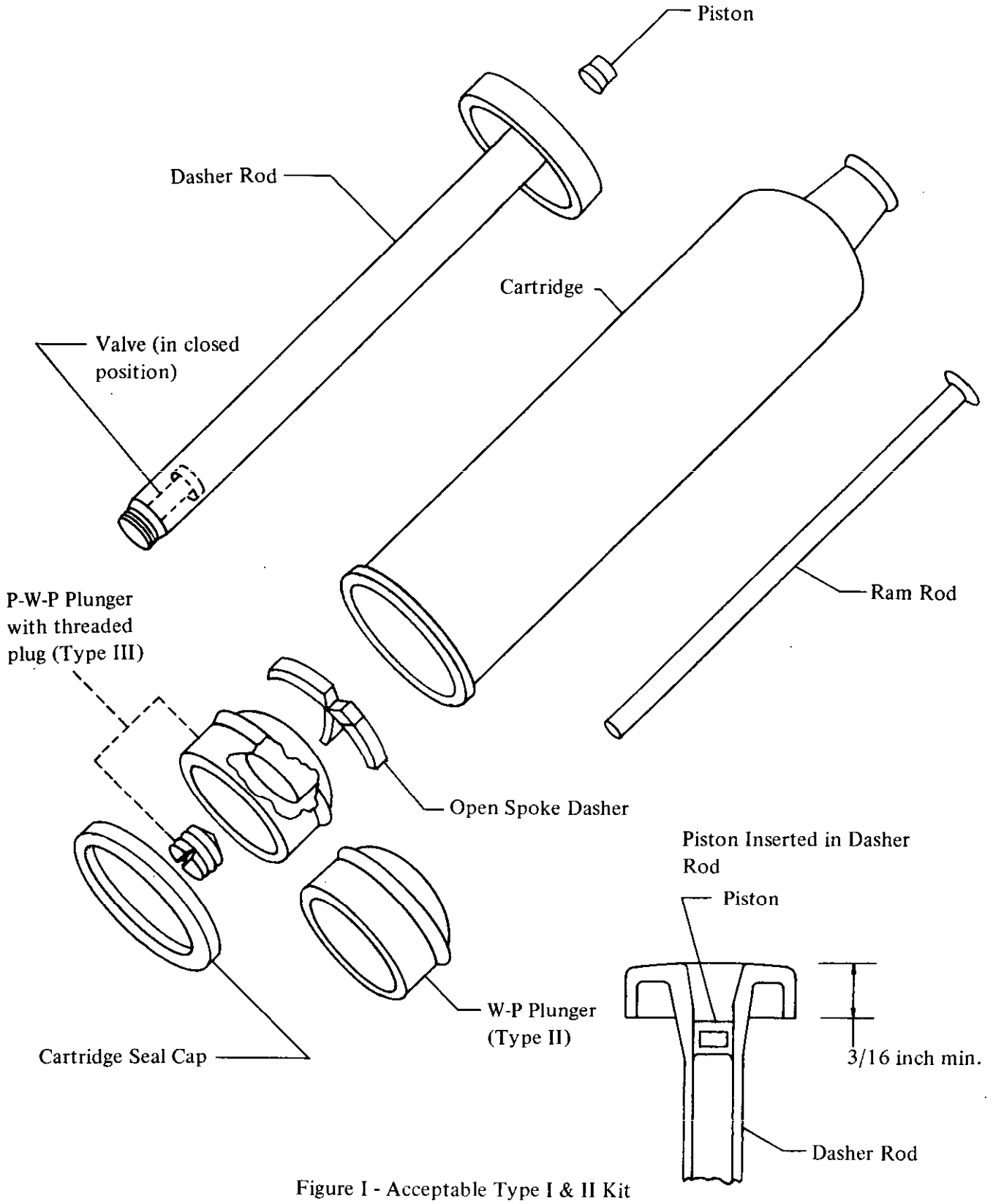


Figure I - Acceptable Type I & II Kit

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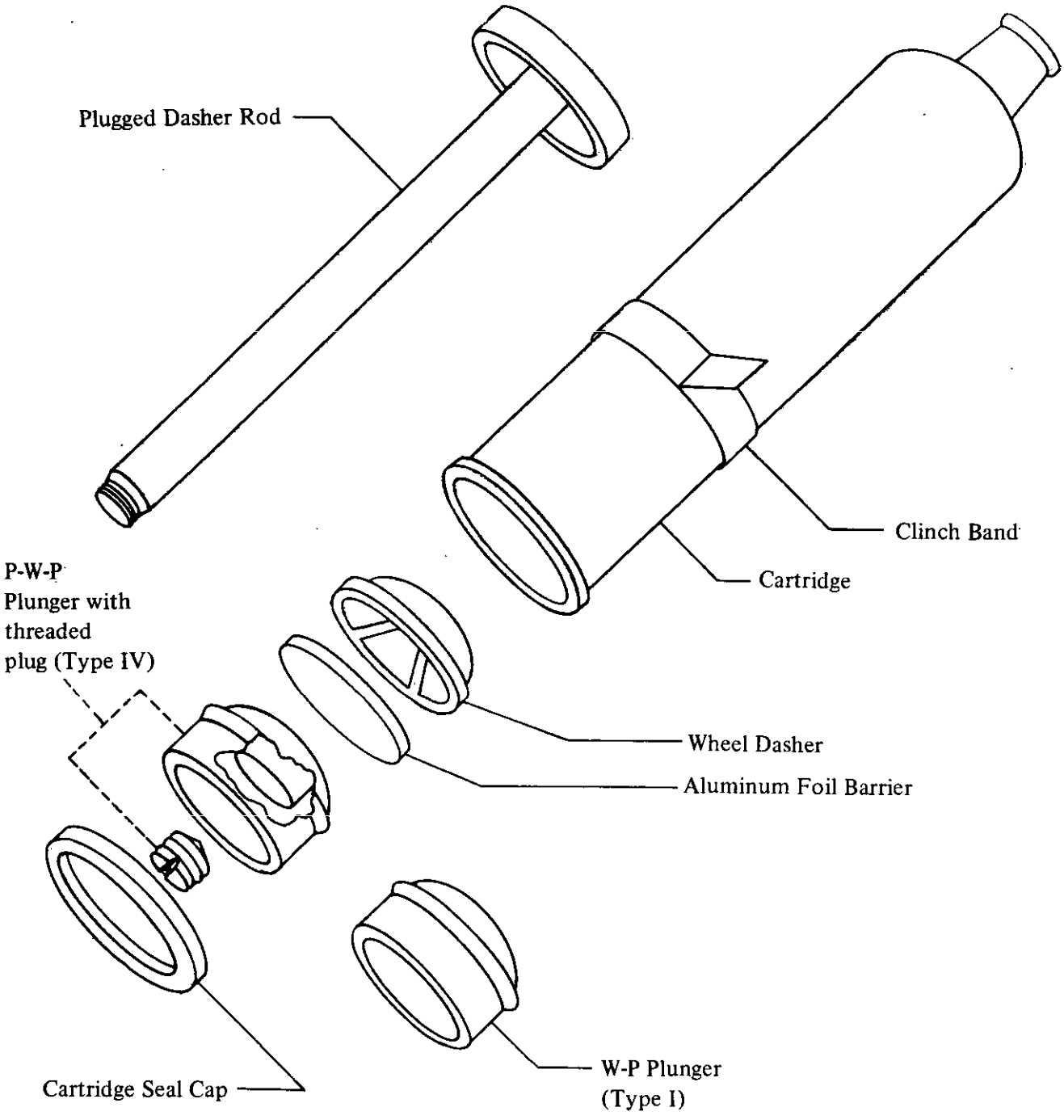


Figure II - Acceptable Type II & IV Kit

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3.2.4 Type IV Cartridge. The Type IV Cartridge shall be as specified for the Type II Cartridge with the addition of a plunger with threaded plug (reference item: Semco P/N 250 PWP) for adaptation to injection guns.

3.3 Filled Cartridge Kit

3.3.1 Material contained in filled kit shall be free of dirt, lumps, chips, or any other foreign material. Type I and III kit shall contain no entrapped air in the dasher rod (catalyst component) or in the cartridge (resin component) except a negligible amount (3%) adjacent to the plunger. Type II and IV kit shall contain no entrapped air in or around the base or catalyst component except a negligible amount (3% maximum) adjacent to the barrier or plunger.

3.3.2 Materials. All materials provided in sealant kits shall conform to Government Specifications, if such specifications exist. Material not governed by a specification shall be of the best quality and conform to requirements of the manufacturer's data sheet.

3.3.3 Cartridge Reliability. The cartridge must work smoothly and the material be completely and correctly mixed when the cartridge instructions are followed.

3.3.4 Bagging. The kit assembly includes a heat-sealed bag into which each filled cartridge is sealed. The bag material shall be mylar-lined aluminum foil or polyethylene, as specified in the contract or ordering data. Unless otherwise specified, mylar-lined aluminum foil shall be used to bag polysulfide, Class A (brushable grade) sealant. Bagging is not required for cartridges which are purchased empty for filling by the user activity.

3.3.5 Temperature Stability. The kit must meet all the requirements of paragraphs 4.3.2.1 and 4.4.5 of this specification after being exposed to 120°F for 24 hours in a forced convection oven.

3.3.6 Long Term Stability. Filled kit suppliers are required to meet the long term storage stability requirements in material specification for the sealant or similar intended product.

3.3.7 Marking and Instructions

3.3.7.1 Kit Marking. Each kit shall have the following information either heat impregnated or screen printed on the side of the cartridge or printed on a label firmly affixed to the side of the cartridge. Letters should be 1/16 inch minimum size arranged in the most practical manner:

a. Material specification number.

b. Class and type (where applicable).

c. Product designation.

d. Batch number.

e. Health precautions.

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3.3.7.1.1 Precautionary Labels. Each kit of sealant or other material shall be marked with the appropriate precautionary labels in accordance with OSHA, EPA and general health practices, and the applicable military material specification. The labels shall be either heat impregnated or screen printed on the side of the cartridge or printed on a label firmly affixed to the side of the cartridge. Letters shall be boldface 1/16 inch minimum size.

3.3.7.1.2 The cartridge should be labeled in bold face letter at least 1/10" size with:

WARNING: Do not reuse container.

3.3.7.1.3 Mixing Instructions. Mixing instructions for both hand and machine mixing, appropriate for material and type of kit shall be on the bag in 1/32" minimum size letters. The instruction shall include the number of strokes for hand mixing and the minutes for machine mixing.

3.4 Reclaimed Materials. Materials reclaimed from solid waste which have been processed to produce an alternate source of raw materials may be used to manufacture new product. None of this shall be construed to mean used or reconstituted product.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for Compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Lot. For examination of preparation for delivery (paragraph 4.3, 4.3.1.1) and visual examination of cartridges (paragraph 4.3.2.1), a lot shall consist of all kits from one manufacturer to be processed at one time. For testing of cartridges (paragraph 4.4.3), a lot shall consist of all cartridges from one manufacturer's batch of material to be processed at one time.

4.3 Sampling and Inspection for Acceptance. Sampling and inspection shall be performed in accordance with MIL-STD-105 except where otherwise indicated herein.

4.3.1 Inspection of Components and Materials. In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used were manufactured, tested and inspected in accordance with the specified requirements of referenced subsidiary specifications and standards and in accordance with this specification, as applicable.

4.3.1.1 Examination of Preparation for Delivery. An examination shall be made to determine that packaging, packing and marking as required by Section 5 of this specification are complied with. Defects shall be scored in accordance with Table I. The sample unit shall be one shipping container and the contents fully prepared for delivery. The lot size shall be expressed in units of shipping containers contained in the inspection lot. The inspection level for this examination is S-4 and the acceptable quality level (AQL) 4.0.

TABLE I

EXAMINATION OF PREPARATION FOR DELIVERY

EXAMINATION	DEFECT
Marking - shipping container interior and intermediate container as applicable.	Incorrect, omitted, illegible, improper, size, location, or application.
Materials	Nonconforming as to type, grade or other category of this specification. Required component missing, damaged or otherwise defective as to affect serviceability.
Workmanship and Preservation	Inadequate packing/packaging of shipping containers. Such as poor closure of container flaps, bulging of containers, insufficient blocking and supports.
Contents	Not as specified or marked.
Weight	Gross or net weight in excess of box certificate weight maximum.

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4.3.2 Inspection of Kits.

4.3.2.1 Visual Examination. The cartridges shall be examined visually for defects in appearance construction, assembly and general workmanship. Visual examination of the cartridges shall be in accordance with the classification of defects set forth in Table II. The inspection level for this examination shall be S-4 and acceptable quality level (AQL) 4.0. The sample unit shall be one cartridge and the lot size expressed in units of cartridges.

4.3.3 Testing of Cartridges. Unless otherwise specified by the contract or purchase order, the supplier shall furnish certified laboratory test reports of the characteristics listed in Table III on each lot of cartridges. The average results applicable to a lot average requirement shall meet the requirements and the number of failures of sample units for characteristics applicable to the sample unit shall not exceed the acceptable quality level. The inspection shall be a S-2 and acceptable quality level (AQL) 1.5. The sample unit shall be one kit and the lot size expressed in units of kits.

TABLE II

VISUAL EXAMINATION

EXAMINATION	DEFECT
Seal	Material leakage around valve, pass plunger, piston seal, cartridge seal, barrier foil, or between dasher rod and cartridge. Material leakage through pin holes, punctures, cuts, etc., in cartridge dasher rod.
Contents	Not as specified or marked. Excessive entrapped air in or around the catalyst or resin materials. (Refer to para 3.3.1) Dirt, metal chips, or any foreign material in the catalyst, resin or container.
Components	Incorrect, or missing components. Seals incorrectly located.
Appearance	Resin, catalyst, dirt soil, ink smears, etc., on exterior of any kit compounds.
Preservation	Rips, punctures, tears, pinholes or cuts in laminated bag. Wrinkles, folds, blisters or holes in the heat seal. Insufficient width of heat sealed area.
Marking and Instructions	Incorrect, omitted, illegible, improper size, location, or application. Precautionary label missing or illegible.
Chemical Attack	Etched areas, cracks, softening, swelling, or any obvious chemical attack on the container assembly.

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TABLE III

TESTING OF KITS

CHARACTERISTICS	RQMTS.	TEST METHOD	DETERMINATION PER UNIT	REPORT TOLERANCE
Working Properties	3.3.3	4.4.2	1	Functional Failure
Temperature Stability	3.3.5	4.4.3	1	----

4.4 Test Methods

4.4.1 Conditioning. The conditioning period prior to test shall be 24 hours. The atmospheric conditions unless otherwise specified shall be 25° ± 1°C (77° ± 2°F) and 50 + 5% relative humidity.

4.4.2 Working Properties. The kit shall be hand mixed in accordance with the attached mixing instructions. The kit shall function correctly including valve opening (Type I and III), complete injection of catalyst into the resin, complete mixing in the specified strokes, proper removal of seal cap and dasher rod and satisfactory extrusion of sufficient mixed material.

4.4.3 Temperature Stability. The assembled kit shall be placed in a forced convection oven which has been preheated to 49 ± 2°C (120± 5°F). After 24 hours the kit shall be removed, allowed to cool to room temperature and subjected to the inspection criteria specified in paragraph 4.3.2.1 and the working properties requirements of paragraph 4.4.4.

5. PACKAGING

5.1 Preservation and Packaging. Preservation and packaging shall be level A and C as specified (see 6.2).

5.1.1 Level-A. The completed retail kits, bagged and marked in accordance with Section 3, shall be packaged in a close-fitting box conforming to PPP-B-636. Each cartridge shall be inserted in a full height cell formed by slotted fiberboard or polystyrene partitions. The box shall be closed in accordance with the appendix to the box specification. Packaging density shall be 24 kits per container.

5.1.2 Level C. The completed retail kits, bagged and marked in accordance with Section 3, shall be packaged in accordance with the supplier's commercial practice.

5.2 Packing. Packing shall be Level A or Commercial as specified (see 6.2).

5.2.1 Level A. Packages prepared in accordance with 5.1 shall be packed in a box conforming to PPP-B-636, class weather resistant. The box shall be closed and strapped in accordance with the appendix to the box specification. The gross weight of the box shall not exceed the weight limitations of the box specification.

5.2.2 Level C. Packages prepared in accordance with 5.1 shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with rules and regulations applicable to mode of transportation.

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5.3 Marking of Shipment. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. Marking shall include specification number, applicable class and dash number, name of manufacturer, date of manufacture (month and year), batch number, mixing instructions, and recommended storage temperature.

6. NOTES

6.1 Intended Use. This specification is intended to control the quality of completely packaged cartridges for two component materials (catalyst and resin components). The cartridges are used for storing, mixing, and applying sealants, adhesives and other similar materials.

6.2 Ordering-Data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

a. Title, number, and date of specification for the packaged material.

b. Type, grade, and class of the packaged material or applicable designation to define the material (i.e., manufacturer's part numbers, formulation numbers, etc).

c. Type and size of cartridge (see 1.2).

d. Amount of material in cartridge (by volume or weight).

e. Style of bag (see 3.3.4).

f. Selection of applicable levels of packaging and packing (see 5).

6.3 Cataloging. It is expected that only Types V and VI will be cataloged into the Federal Supply System under this specification and part number system. Federal Stock Class (FSC) 5120 applies for the empty cartridges. Filled cartridges will be cataloged under the applicable material specification in FSC 8030 or other as appropriate.

6.4 Changes From Previous Revision. Due to extensiveness of changes, asterisks could not have been used to mark changes from previous revision.

Custodian:

Air Force - 99

Review Activities:

Air Force - 85

Preparing Activity:

Air Force - 84

Project Number:

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions — Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-38714B	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rational for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* - Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* - Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* - Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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